



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

ASSIGNMENT 2 – SPRING 2026

COURSE NAME:

Multimedia Application

COURSE NUMBER:

SOC4080

READ INSTRUCTIONS FIRST:

Any form of cheating is strictly prohibited.

Write your Name (*capital letters*), ID number and Section number on each page.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use a correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

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Part 1. Write your Statement of Purpose (SOP)

7.5

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Part 2. Calculate One Pass of Self-Attention.

[/7.5]

Token embeddings matrix:

Fixed projection matrices:

$$W^Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad W^K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W^V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Step 1. Compute Q, K, V

[/3]

$$Q = X W^{Q\top}, \quad K = X W^{K\top}, \quad V = X W^{V\top}.$$

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Step 2. Form the Score Matrix S

[/1]

$$S = Q K^{\top}, \quad S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

Step 3. Scale the Scores.

$$\tilde{S} = \frac{S}{\sqrt{d_k}} \quad \text{with } d_k = 2,$$

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Step 4. Convert to Attention Weights

$$A = \text{softmax}(\tilde{S}) \quad (\text{apply softmax **row-wise**}).$$

Step 5. Compute the Output Z

$$Z = AV, \quad Z_i = \sum_{j=1}^2 A_{ij} \mathbf{v}_j,$$